⇑Sagar Panchal
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About

Senior engineer specializing in supporting and optimizing complex automation and robotic systems in production environments. I combine software expertise, motion control, machine vision, and real-time diagnostics to troubleshoot, stabilize, and enhance high-precision equipment. Focused on roles requiring deep technical ownership, on-site problem solving, and end-to-end delivery of reliable machine-level solutions.

Education

MS, Computer engineering (3.41/4) – Dec-17
New York University NYU
BE, Instrumentation & Controls Engineering (7.04/10) – Jun-14
Gujarat Technological University

Skills

Automation:	Machine Automation - PC based Systems, Industrial Automation - PLC/HMI based Systems, Embedded, Distributed & Real Time Systems (VxWorks) Computer Vision (Cognex, Keyence, OpenCV) & Image Processing
Software:	C, CPP, CS, Java, Ladder, FBD, Web - HTML, CSS, JS SCM : Git (Bash & Sourcetree) AI driven workflows - Python, Swift

Professional Experience

Software Engineer III - Automation, Universal Instruments, NY — *Sep 22 - present*

Core Responsibilities:

- Maintain and evolve high-precision, high-speed software modules across the Fuzion SMT platform, ensuring reliability, performance, and production readiness.
- Lead remote production-line diagnostics by analyzing logs, replicating site conditions, performing root-cause investigations and delivering targeted fixes, patches, and release updates.
- Conduct code reviews and enforce software-quality standards across the platform, ensuring robust design, maintainability and consistency in mission-critical automation workflows.
- Enable field and application engineers to successfully deploy and validate applications by providing technical guidance, interpreting machine data, and supplying required configurations or software updates.

Key contributions:

- Delivered Selective Placement, Fuzion's first smart pick/buffer/place workflow using DFF-based categories and avoiding-priority rules, improving product quality and placement intelligence.
- Engineered SBS using Keyence SR-X300, delivering Fuzion's first side-barcode scanning capability with live-view, traceability, and full error-recovery support.
- Upgraded the AMS workflow from a single point to 13 wafer-level locations, significantly improving placement accuracy and calibration robustness.
- Built the wafer-to-board mapping system that converts customer wafer maps into Fuzion board templates, eliminating hours of manual template creation and enabling direct wafer-data usage.
- Implemented AVP, enabling user-defined fiducial finds across ECS, NPI, and Production workflows, increasing alignment flexibility and reducing setup constraints.
- Integrated Cognex QuickBuild (VPP) with MCOS↔QBIT TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Precision-Bonder platform for the first time.
- Delivered multiple platform-level enhancements including the Light Intensity Controller, nested-tray support, isosceles fiducial and configurable park-position, strengthening core system capability.

Automation Engineer , Nor-Cal controls, Ca — *Jan - Jul 22*

- Designed SCADA interfaces and integrated PLC logic with inverter and sensor data to deliver reliable solar power-plant automation including FAT and commissioning support.
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

Software Engineer - Machine Automation, Mini-Circuits, NY — *Feb 18 - Nov 21*

- Evolved automation software across rotary machines coordinating servo-motion via PCI-e, sensor-driven logic, CV inspection, EVT automation, and database-supported workflows.
- Enhanced precision laser-measurement stations, semiconductor-packaging machines, and PID-controlled TVAC systems by integrating database-linked classification logic, improving QA reliability and reducing manual effort.
- Built and modernized full-stack systems including the public shopping-cart redesign, product-visibility management and lifecycle-tracking tools, increasing platform reliability and removing hours of manual data handling.

Developer Intern, RevMax Fleet Optimization, NY based startup — *Fall 2017*

- Developed Doppler-radar based pedestrian-counting device and transferred data to back-end services using BLE.
- Deployed sensors at multiple NYC locations and validated field performance

Instructor (ITEST Program), robotics & mechatronics dpt., NYU — *Fall 2017*

- Mentored high-school robotics teams in autonomous-robot workflows for FIRST-Robotics/VEX competitions.
- Taught foundational programming concepts in Arduino & sensor-integration to student robotics teams.

Lead Instructor (CrEST Program), Center for K12 STEM Education, NYU — *Summer 2017*

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
  -Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
  -Wireless Robot Controller - Arduino: https://github.com/sap586/WirelessRobotDay5Activity

Automation Engineer, Prima Automation, Ahmedabad, IN — *Jun 14 - Jan 16*

- Engineered machine-automation workflows using Siemens PLCs & HMI interfaces (TIA Portal) integrating motion control, sensors and industrial communication protocols across full design-to-deployment cycles.
- Designed and validated electrical control systems - AutoCAD panel layouts, E-plan schematics, site installation, and FAT support - ensuring reliable integration between software, motion hardware, and production robotics.

This document is prepared using LaTeX. Click here to download the source code.